

Yeshwanth Cherapanamjeri

Nanyang Assistant Professor

CONTACT INFORMATION	Nanyang Technological University College of Computing and Data Science	https://yeshwanth94.github.io yesh.c@ntu.edu.sg
INTERESTS	Algorithms, Statistical Learning Theory, Optimization	
EDUCATION	UC Berkeley Ph.D Student in Computer Science Advisor: Prof. Peter L. Bartlett CGPA: 4.0+	(August 2017 - August 2023)
	Indian Institute of Technology Bombay B. Tech with Honors in Computer Science and Engineering Minor in Applied Statistics and Informatics CGPA: 9.31 (<i>Ranked among the top 10% of the department</i>)	(July 2011 - May 2015)
EMPLOYMENT	Nanyang Technological University <i>Nanyang Assistant Professor</i>	(March 2026 - Present)
	Massachusetts Institute of Technology <i>Postdoctoral Associate</i>	(September 2023 - September 2025) Advisor: Prof. Constantinos Daskalakis
	The Voleon Group <i>Research Scientist Intern</i>	(May 2021 - August 2021) Manager: Dr. Neal Master
	Amazon Inc <i>Applied Scientist Intern</i>	(June 2020 - August 2020) Managers: Dr. Choon Hui Teo and Dr. Vishy Vishwanathan
	Microsoft Research India <i>Research Fellow</i>	(June 2015 - July 2017) Advisors: Dr. Prateek Jain and Dr. Praneeth Netrapalli
	TU Braunschweig <i>Research Intern</i>	(May 2013 - July 2013) Advisor: Prof. Marcus Magnor
SELECTED PUBLICATIONS	Statistical Limits of Affine Equivariant Estimation Z. Chen, Y. Cherapanamjeri Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026) ArXiv Version: https://arxiv.org/abs/2310.10758	
	Learning Correlated Reward Models: Statistical Barriers and Opportunities Y. Cherapanamjeri*, C. Daskalakis, G. Farina, S. Mohammadpour* Fourteenth International Conference on Learning Representations (ICLR 2026) ArXiv Version: https://arxiv.org/abs/2510.15839	
	How Much is a Noisy Image Worth? Data Scaling Laws for Ambient Diffusion G. Daras*, Y. Cherapanamjeri*, C. Daskalakis Thirteenth International Conference on Learning Representations (ICLR 2025) ArXiv Version: https://arxiv.org/abs/2411.02780	
	Optimal PAC Bounds without Uniform Convergence Sixty Fourth Symposium on Foundations of Computer Science (FOCS 2023) <i>Invited to SICOMP Special Issue for FOCS 2023</i> ArXiv Version: https://arxiv.org/abs/2304.09167	
	What Makes A Good Fisherman? Linear Regression under Self-Selection Bias Y. Cherapanamjeri, C. Daskalakis, A. Ilyas, E. Zampetakis Fifty Fifth Symposium on Theory of Computing (STOC 2023) ArXiv Version: https://arxiv.org/abs/2205.03246	

Estimation of Standard Auction Models

Y. Cherapanamjeri, C. Daskalakis, A. Ilyas, E. Zampetakis

Extended Abstract: Twenty Third Conference on Economics and Computation (EC 2022)

ArXiv Version: <https://arxiv.org/abs/2205.02060>

Algorithms for Heavy-Tailed Statistics: Regression, Covariance Estimation, and Beyond

Y. Cherapanamjeri, S. B. Hopkins, T. Kathuria, P. Raghavendra, N. Tripuraneni

Fifty Second Symposium on Theory of Computing (STOC 2020)

ArXiv Version: <https://arxiv.org/abs/1912.11071>

Fast Mean Estimation with Sub-Gaussian Rates

Y. Cherapanamjeri, N. Flammarion, P. L. Bartlett

Thirty Second Conference on Learning Theory (COLT 2019)

ArXiv Version: <https://arxiv.org/abs/1902.01998>

PUBLICATIONS

Phase Transitions in Heavy-Tailed Mean Estimation under ℓ_p Norms

I. Aden-Ali, Y. Cherapanamjeri, M. M. Høgsgaard, K. G. Larsen, N. Zhivotovskiy

Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026)

Statistical Limits of Affine Equivariant Estimation

Z. Chen, Y. Cherapanamjeri

Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026)

ArXiv Version: <https://arxiv.org/abs/2310.10758>

Learning Correlated Reward Models: Statistical Barriers and Opportunities

Y. Cherapanamjeri*, C. Daskalakis, G. Farina, S. Mohammadpour*

Fourteenth International Conference on Learning Representations (ICLR 2026)

ArXiv Version: <https://arxiv.org/abs/2510.15839>

How Much is a Noisy Image Worth? Data Scaling Laws for Ambient Diffusion

Thirteenth International Conference on Learning Representations (ICLR 2025)

ArXiv Version: <https://arxiv.org/abs/2411.02780>

Heavy-tailed Estimation is Easier than Adversarial Contamination

Y. Cherapanamjeri, D. Lee

Thirty Eighth Conference on Learning Theory (COLT 2025)

ArXiv Version: <https://arxiv.org/abs/2411.15306>

Computing Approximate Centerpoints in Polynomial Time

Y. Cherapanamjeri

Sixty Fifth Symposium on Foundations of Computer Science (FOCS 2024)

The Space Complexity of Learning-Unlearning Algorithms

Y. Cherapanamjeri, S. Garg, N. Rajaraman, A. Sekhari, A. Shetty

Thirty Eighth Conference on Learning Theory (COLT 2025)

Efficient Automated Circuit Discovery in Transformers using Contextual Decomposition

A. Hsu, G. Zhou, Y. Cherapanamjeri, Y. Huang, A. Odisho, P. Carroll, B. Yu

Thirteenth International Conference on Learning Representations (ICLR 2025)

ArXiv Version: <https://arxiv.org/abs/2407.00886>

Diagnosing Transformers: Illuminating Feature Spaces for Clinical Decision-Making

A. R. Hsu, Y. Cherapanamjeri, B. Park, T. Naumann, A. Y. Odisho, B. Yu

Twelfth International Conference on Learning Representations (ICLR 2024)

ArXiv Version: <https://arxiv.org/abs/2305.17588>

Optimal PAC Bounds without Uniform Convergence

Sixty Fourth Symposium on Foundations of Computer Science (FOCS 2023)

Invited to SICOMP Special Issue for FOCS 2023

ArXiv Version: <https://arxiv.org/abs/2304.09167>

The One-Inclusion-Graph Algorithm is not Always Optimal

I. Aden-Ali, Y. Cherapanamjeri, A. Shetty, N. Zhivotovskiy

Thirty Sixth Conference on Learning Theory (COLT 2023)

ArXiv Version: <https://arxiv.org/abs/2212.09270>

Optimal Algorithms for Linear Algebra in the Current Matrix Multiplication Time

Y. Cherapanamjeri, S. Silwal, D. P. Woodruff, S. Zhou
ACM-SIAM Symposium on Discrete Algorithms (SODA 2023)
ArXiv Version: <https://arxiv.org/abs/2211.09964>

Robust Algorithms on Adaptive Inputs from Bounded Adversaries

Y. Cherapanamjeri, S. Silwal, D. P. Woodruff, F. Zhang, Q. Zhang, S. Zhou
Eleventh International Conference on Learning Representations (ICLR 2023)
ArXiv Version: <https://arxiv.org/abs/2304.07413>

What Makes A Good Fisherman? Linear Regression under Self-Selection Bias

Y. Cherapanamjeri, C. Daskalakis, A. Ilyas, E. Zampetakis
Fifty Fifth Symposium on Theory of Computing (STOC 2023)
ArXiv Version: <https://arxiv.org/abs/2205.03246>

Estimation of Standard Auction Models

Y. Cherapanamjeri, C. Daskalakis, A. Ilyas, E. Zampetakis
Extended Abstract: Twenty Third Conference on Economics and Computation (EC 2022)
ArXiv Version: <https://arxiv.org/abs/2205.02060>

Uniform Approximations for Randomized Hadamard Transforms with Applications

Y. Cherapanamjeri, J. Nelson
Fifty Fourth Symposium on Theory of Computing (STOC 2022)
ArXiv Version: <https://arxiv.org/abs/2203.01599>

Adversarial Examples in Multi-Layer Random ReLU Networks

P. L. Bartlett, S. Bubeck, Y. Cherapanamjeri
Thirty Fifth Conference on Neural Information Processing Systems (NeurIPS 2021)
ArXiv Version: <https://arxiv.org/abs/2106.12611>

A single gradient step finds adversarial examples on random two-layers neural networks

S. Bubeck, Y. Cherapanamjeri, G. Gidel, R. Tachet des Combes
Thirty Fifth Conference on Neural Information Processing Systems (NeurIPS 2021)
Spotlight Presentation
ArXiv Version: <https://arxiv.org/abs/2104.03863>

Terminal Embeddings in Sublinear Time

Y. Cherapanamjeri, J. Nelson
Sixty Second Symposium on Foundations of Computer Science (FOCS 2021)

On Adaptive Distance Estimation

Y. Cherapanamjeri, J. Nelson
Thirty Fourth Conference on Neural Information Processing Systems (NeurIPS 2020)
Spotlight Presentation
ArXiv Version: <https://arxiv.org/abs/2010.11252>

Optimal Robust Linear Regression in Nearly Linear Time

Y. Cherapanamjeri, E. Aras, N. Tripuraneni, M. I. Jordan, N. Flammarion, P. L. Bartlett
In Submission
ArXiv Version: <https://arxiv.org/abs/2007.08137>

List Decodable Mean Estimation in Nearly Linear Time

Y. Cherapanamjeri, S. Mohanty, M. Yau
Sixty First Symposium on Foundations of Computer Science (FOCS 2020)
ArXiv Version: <https://arxiv.org/abs/2005.09796>

Optimal Mean Estimation without a Covariance

Y. Cherapanamjeri, N. Tripuraneni, P. L. Bartlett, M. I. Jordan
In Submission

Algorithms for Heavy-Tailed Statistics: Regression, Covariance Estimation, and Beyond

Y. Cherapanamjeri, S. B. Hopkins, T. Kathuria, P. Raghavendra, N. Tripuraneni
Fifty Second Symposium on Theory of Computing (STOC 2020)
ArXiv Version: <https://arxiv.org/abs/1912.11071>

Fast Mean Estimation with Sub-Gaussian Rates

Y. Cherapanamjeri, N. Flammarion, P. L. Bartlett

Thirty Second Conference on Learning Theory (COLT 2019)

ArXiv Version: <https://arxiv.org/abs/1902.01998>**Testing Markov Chains without Hitting**

Y. Cherapanamjeri, P. L. Bartlett

Thirty Second Conference on Learning Theory (COLT 2019)

ArXiv Version: <https://arxiv.org/abs/1902.01999>**Thresholding based Efficient Outlier Robust PCA**

Y. Cherapanamjeri, P. Jain, P. Netrapalli

Thirtieth Conference on Learning Theory (COLT 2017)

ArXiv Version: <https://arxiv.org/abs/1702.05571>**Nearly Optimal Robust Matrix Completion**

Y. Cherapanamjeri, K. Gupta, P. Jain

Thirty-Fourth International Conference on Machine Learning (ICML 2017)

ArXiv Version: <https://arxiv.org/abs/1606.07315>**SELECTED
TALKS****Statistical Challenges in Modern Machine Learning
and their Algorithmic Consequences**

Spring 2025

- Carnegie Mellon University
- Nanyang Technological University
- Rutgers University
- University of Maryland
- University of Texas at Austin
- University of Waterloo
- University of Wisconsin-Madison

Affine Equivariant Robust Mean Estimation

WALE 2024

Optimal PAC Bounds without Uniform Convergence

A&C Seminar MIT 2023

Recovering from Structured and Unstructured Noise

Harvard 2023

Uniform Approximations for RHTs

Stanford Theory Lunch 2022

Towards Adaptive Metric Embeddings

Metric Embeddings Workshop FOCS 2022

Optimal Mean Estimation without a Variance

Virtual Workshop 2022

Optimal Mean Estimation without a Variance

Virtual CS4Math Seminar Harvard 2021

A General Framework for Adaptive Data Structures

STOC 2021

Workshop on Robust Streaming, Sketching, and Sampling

A General Framework for Adaptive Data Structures

Google Research 2021

**SELECTED
AWARDS**

Invitation to SICOMP Special Issue for FOCS 2023

2023

Finalist: Two Sigma Graduate Fellowship

2021

Outstanding Graduate Student Instructor

2018

TEACHING	EECS 127/227A: Optimization Models in Engineering , UC Berkeley	Spring 2020
	<i>Instructor: Prof. Gireeja Ranade</i>	
	Graduate Student Instructor	
	CS 170: Efficient Algorithms and Intractable Problems , UC Berkeley	Spring 2019
	<i>Instructors: Prof. Prasad Raghavendra and Prof. Luca Trevisan</i>	
	Graduate Student Instructor	
	CS 70: Discrete Mathematics and Probability Theory , UC Berkeley	Fall 2018
	<i>Instructors: Prof. Alistair Sinclair and Prof. Yun Song</i>	
	Graduate Student Instructor	
	<i>Outstanding GSI Award</i>	
	MA 214: Introduction to Numerical Analysis , IIT Bombay	Summer 2014
	<i>Instructor: Prof. Sivaji Ganesh</i>	
	Undergraduate Student Instructor	
PROFESSIONAL SERVICE	Conference Reviewing: STOC 2025, ALT 2025, NeurIPS 2024, STOC 2024, NeurIPS 2023, FOCS 2023, ICML 2023, STOC 2023, ICLR 2023, NeurIPS 2022, STOC 2022, FOCS 2021, NeurIPS 2021, ICML 2021, STOC 2021, FOCS 2020, ICML 2019, COLT 2019, SODA 2019	
	Journal Reviewing: Annals of Statistics, Bernoulli, Journal of the ACM (JACM), Transactions on Information Theory, Mathematical Statistics and Learning, Journal of Machine Learning Research (JMLR)	